To comprehensively cover the topics outlined in the BCA-102 Programming Principle Algorithm syllabus, here are some curated YouTube videos and online resources:

Unit I: Introduction to C Language

- **YouTube Video:** "C Programming Tutorial for Beginners"
  - This video provides a full introduction to core concepts in C programming, including its history, structure, and functions.
- **Online Resource:** "C Programming Language Tutorial - GeeksforGeeks"
  - This tutorial covers language fundamentals such as character sets, tokens, keywords, identifiers, variables, constants, data types, and comments.

Unit II: Operators

- **YouTube Video:** "C Programming Full Course for Beginners"
  - This comprehensive course includes sections on operators, expressions, statements, and console-based I/O functions like printf() and scanf().
- **Online Resource:** "C Tutorial - W3Schools"
  - This resource provides detailed explanations of operator types, precedence, associativity, and preprocessor directives such as #include and #define.

Unit III: Control Structures

- **YouTube Playlist:** "C Programming Tutorials"
  - This playlist offers tutorials on decision-making structures (if, if-else, switch) and loop control structures (while, do-while, for), including the use of break, continue, goto, and exit statements.
- **Online Resource:** "Learn C Programming - Programiz"
  - This platform provides interactive lessons on control structures with practical examples.

Unit IV: Introduction to Problem Solving

- **YouTube Video:** "Algorithm and Flowchart"
  - This video explains problem-solving techniques, steps in problem-solving, and the creation of algorithms and flowcharts.
- **Online Resource:** "Introduction to Flowcharts - GeeksforGeeks"
  - This article discusses the symbols and characteristics of algorithms and flowcharts, including conditionals and loops in pseudocode.

Unit V: Simple Arithmetic Problems

- **YouTube Video:** "Algorithms & Flowcharts"

    - This tutorial covers algorithms and flowcharts for arithmetic problems such as addition, multiplication, determining positive/negative numbers, even/odd checks, and more.

- **Online Resource:** "Explain Algorithm and Flowchart with Examples - Edraw"
    - This resource provides examples of algorithms and flowcharts for various arithmetic problems, including GCD and Pascal's Triangle.

Unit VI: Functions

- **YouTube Video:** "C Programming Full Course for Free"
    - This full course includes detailed sections on functions, covering declaration, definition, function calls, parameter passing, scope of variables, storage classes, and recursion.
- **Online Resource:** "C Tutorial - TutorialsPoint"
    - This tutorial offers in-depth information on functions in C, including call by value, call by reference, and recursive functions.

These resources should provide a comprehensive understanding of each unit in your BCA-102 syllabus.